(1) Funcții olomorfe

Relațiile Cauchy-Riemann

$$\frac{\partial P}{\partial x}(x_0, y_0) = \frac{\partial Q}{\partial y}(x_0, y_0)$$
$$\frac{\partial P}{\partial y}(x_0, y_0) = -\frac{\partial Q}{\partial x}(x_0, y_0)$$

$$f = P + iQ \Rightarrow \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right)$$

$$\left.\begin{array}{l} f = P + iQ \\ \frac{\partial f}{\partial \bar{z}} = 0 \\ P, Q \in \mathcal{C}^1 \end{array}\right| \Rightarrow f \text{ olomorfă}$$

$$\Delta Q = \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2} = 0 \Rightarrow Q \text{ funcție armonică}$$

f olomorfă $\Rightarrow$ P, Q armonice

$$\left.\begin{array}{l} P \text{ armonică} \\ \exists\, Q \text{ armonică} \end{array}\right\} \text{a.î. } f \text{ olomorfă}$$

(2) Functii

$$e^x = \sum_{n \geq 0} \frac{z^n}{n!}$$

$$w = u + iv = \ln z = \{\ln|z| + i(\arg z + 2k\pi);\ k \in \mathbb{Z}\}$$

$$z = r(\cos\theta + i\sin\theta) \qquad \theta = \arg z$$

$$z^w = \{|z|^w \cdot e^{iw(\arg z + 2k\pi)},\ k \in \mathbb{Z}\}$$

$$\sqrt[n]{z} = \left\{\sqrt[n]{|z|} \cdot e^{i\frac{\arg z + 2k\pi}{n}};\ k \in \overline{0, n-1}\right\}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\operatorname{sh} z = \frac{e^z - e^{-z}}{2} \qquad \operatorname{ch} z = \frac{e^z + e^{-z}}{2}$$

$$\sin iz = i\operatorname{sh} z$$

$$\cos iz = \operatorname{ch} z$$

(3) Integrale complexe

$$\int_\gamma f(z)\,dz = \int_\gamma P\,dx - Q\,dy + i\int_\gamma Q\,dx + P\,dy$$

$$= \int_a^b f(z(t))\cdot z(t)'\,dt \quad \text{unde } \gamma = \gamma(t) \;;\; t\in[a,b]$$

$$z(t) = x(t) + i\,y(t)$$

Parametrizare cerc $\quad \gamma: z(t) = z_0 + Re^{it} \quad t\in[0,2\pi]$

Parametrizare segment (de la z_1 la z_2) $\quad z(t) = (1-t)z_1 + tz_2 \quad t\in[0,1]$

Th Cauchy

funcție olomorfă $f = P + iQ$
γ drum închis
se verifică condițiile G-R
$\Rightarrow \int_\gamma f(z)\,dz = 0$

Formula Leibniz-Newton

$$\int_{z_0}^{z_1} f(z)\,dz = F(z_1) - F(z_0)$$

$$F'(z) = f(z)$$

f olomorfă
mărginită de γ închisă
și fără autointersecții

$$\Rightarrow f(a) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-a}\,dz$$

$$f^{(n)}(a) = \frac{n!}{2\pi i}\int_\gamma \frac{f(z)\,dz}{(z-a)^{n+1}}$$

(4) Serii Laurent

$$\sum_{n\geq 0} a_n (z-z_0)^n$$

Raza conv. $R = \dfrac{1}{\lim\limits_{n\to\infty} \sqrt[n]{|a_n|}} = \dfrac{1}{\lim\limits_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right|}$

$|z-z_0| < R \rightarrow$ convergentă

$|z-z_0| > R \rightarrow$ divergentă

Partea principală $\sum\limits_{n\geq 1} a_{-n}(z-z_0)^{-n}$

$$e^z = \sum_{n\geq 0} \frac{z^n}{n!} \quad \forall z \in \mathbb{C}$$

$$\sin z = \sum_{n\geq 0} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad \forall z \in \mathbb{C}$$

$$\cos z = \sum_{n\geq 0} (-1)^n \frac{z^{2n}}{(2n)!} \quad \forall z \in \mathbb{C}$$

$$\frac{1}{1-z} = \sum_{n\geq 0} z^n \qquad |z|<1$$

$$\frac{1}{1+z} = \sum_{n\geq 0} (-1)^n z^n \qquad |z|<1$$

$$(1+z)^\alpha = 1 + \sum_{n\geq 1} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} z^n \qquad |z|<1$$

(5) Puncte singulare

Punct singular aparent

- partea principală a serie este nulă
- $\lim_{z \to z_0} f(z)$ există și e finită

→ Reziduul = 0

Punct singular esențial

- partea principală a serie are o infinitate de termeni nenuli
- $\lim_{z \to z_0} f(z) \nexists$

→ Reziduul → dezvoltarea în serie Laurent cu coeficientul a_{-1} (al lui $\frac{1}{z-z_0}$)

$$a_{-1} = \frac{1}{2\pi i} \int_C f(z)dz$$

$C \to$ arc de centru z_0, rază R

Pol

- partea principală a serie are numărul finit de termeni nenuli
- $\lim_{z \to z_0} f(z) = 0$

$$\text{Rez}(f, z_0) = \frac{1}{(k-1)!} \lim_{z \to z_0} \left((z-z_0)^k \cdot f(z) \right)^{(k-1)}$$

$$\int_{\gamma} f(z)dz = 2\pi i \sum_{j=1}^{k} \operatorname{Rez}(f, \alpha_j)$$

$$\operatorname{Rez}(f, \infty) = \frac{-1}{2\pi i} \int_{C} f(z)dz$$

SAU $$\operatorname{Rez}(f, \infty) = -\sum_{j=1}^{k} \operatorname{Rez}(f, \alpha_j)$$

(6) Tipuri integrale

Tipul I → conține sin sau cos

→ numitorul nu se anulează pe $|z|=1$

→ se face schimbare de variabilă cu $z=e^{it}$

→ se scrie sin și cos în funcție de e^t

Tipul II → $\int_{-\infty}^{\infty} R(x)\,dx$

$R(x)$ funcție reală

$\lim_{|x|\to\infty} x\,R(x)=0$

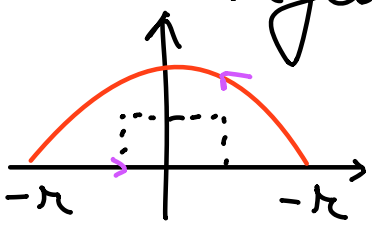
→ se scrie $f(z)$ (înlocuim în funcție x cu z)

→ se află polii

→ în derese se pun polii din $f(z)$

→ Se scrie textul:

Alegem $r>1$ ($\sqrt{2}$ .... etc) a.î. $\gamma=\gamma_r\cup[-r;r]$



$$\int_{\gamma} f(z)\,dz \overset{th.\ Rez}{=} 2\pi i \sum \mathrm{Rez}(f,\alpha_i)$$

$$\int_{\gamma_r} f(z)\,dz + \int_{-r}^{r} f(x)\,dx \quad (1)$$

$\lim_{|z|\to\infty} z\cdot f(z) = \ldots = 0 \overset{\text{Lema Jordan}}{\Longrightarrow} \int_{\gamma_2} f(z)\,dz \xrightarrow[r\to\infty]{} 0$

Înl (1) aplic $r\to\infty \Rightarrow I=\int_{-\infty}^{\infty} f(x)\,dx$

Tipul III → $\int_{-\infty}^{\infty} f(x) e^{ix} dx$

→ f olomorfă

→ Se folosește γ și Γ a.î. $\gamma + i\Gamma$ să se calculeze și la final să se facă identificarea

→ Se parcurg pașii de la T II

(7) Laplace

$$u(t)=\begin{cases}1 & ; t\geq 0\\ 0 & ; t<0\end{cases}$$

$$F(p)=\int_0^{\infty} f(t)\cdot e^{-pt}dt \Rightarrow f(t)\xrightarrow{\mathcal{L}} F(p)$$

$$\mathcal{L}[f(t)](p)=F(p)$$

$$f(t)=\frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty} F(p)\cdot e^{pt}dp$$

$$f(t)=\sum_k \operatorname{Rez}\left(F(p)e^{pt}, p_k\right)$$

$u(t)$	$1/p$
t^u	$\Gamma(u+1)/p^{u+1}$
e^{wt}	$\frac{1}{p-w}$
$\sin(wt)$	$\frac{w}{p^2+w^2}$
$\cos(wt)$	$\frac{p}{p^2+w^2}$
$\operatorname{sh}(wt)$	$\frac{w}{p^2-w^2}$
$\operatorname{ch}(wt)$	$\frac{p}{p^2-w^2}$

$\Gamma(u+1)=u!$

Liniaritatea $\alpha f(t)+\beta g(t) \xrightarrow{\mathcal{L}} \alpha F(p)+\beta G(t)$

Asemănarea $f(t) \xrightarrow{\mathcal{L}} F(t)$

$f(at) \xrightarrow{\mathcal{L}} \frac{1}{a} F\left(\frac{t}{a}\right)$

Întârzierea $f(t-\tau) \xrightarrow{\mathcal{L}} e^{-p\tau} F(p)$

Deplasarea $e^{-\lambda t} f(t) \xrightarrow{\mathcal{L}} F(p+\lambda)$

Derivata imag. $t^n f(t) \xrightarrow{\mathcal{L}} (-1)^n F^{(n)}(p)$

Derivata orig. $f^{(n)}(t) \xrightarrow{\mathcal{L}} p^n F(p) - p^{n-1} f(0_+)$

$- p^{n-2} f'(0_+) + \ldots - f^{(n-1)}(0_+)$

Integrarea orig. $\int_0^t f(\tau)\,d\tau \xrightarrow{\mathcal{L}} \frac{F(p)}{p}$

Integ. imag. $\frac{f(t)}{t} \xrightarrow{\mathcal{L}} \int_p^\infty F(\theta)\,d\theta$

Convolutia

$$(f*g)(t) = \int_0^t f(\tau) g(t-\tau)\,d\tau \xrightarrow{\mathcal{L}} F(p)\cdot G(p)$$

(8) Transformata Z

$$\delta_k = \begin{cases} 1 ; & n = k \\ 0 ; & n \neq k \end{cases}$$

Inversa $$x_n = \frac{1}{2\pi i} \int_{|z|=\rho} L_x(z) \cdot z^{n-1} dz$$

atunci $$x_n = \sum_k \operatorname{Rez}\left(L_x(z) \cdot z^{n-1}, z_k\right)$$

Liniaritatea $$L_{\alpha x + \beta y}(z) = \alpha L_x(z) + \beta L_y(z)$$

Convolutia

$$x * y \in S_d^+$$

$$L_{x*y}(z) = L_x(z) \cdot L_y(z)$$

$$L_{x * \delta_k}(z) = z^{-k} \cdot L_x(x)$$

$$\delta_k(n) \xrightarrow{z} \sum_{n \in \mathbb{Z}} \delta_k(n) \cdot z^{-n} = z^{-k}$$

$$h(n) \xrightarrow{z} \sum_{n \in \mathbb{Z}} h(n) \cdot z^{-n} = \sum_{n \geq 0} z^{-n} = \sum_{n \geq 0} \left(z^{-1}\right)^n = \frac{1}{1 - \frac{1}{z}}$$

$$a^n h(n) \xrightarrow{z} \sum_{n \in \mathbb{Z}} a^n h(n) \cdot z^{-n} = \sum_{n \geq 0} \left(a \cdot z^{-1}\right)^n = \frac{1}{1 - \frac{a}{z}}$$

$$n h(n) \xrightarrow{\mathcal{Z}} \sum_{n\in\mathbb{Z}} n h(n)\cdot z^{-n} = \sum_{n\geq 0} n z^{-n} = -z \sum_{n\geq 0} -n z^{n-1}$$

$$= -z \sum_{n\in\mathbb{Z}} \left(h(n) z^{-n}\right)' = -z \cdot \left(\frac{z}{z-1}\right)'$$

$$a^n x_n h(n) \xrightarrow{\mathcal{Z}} L_x\left(\frac{z}{a}\right)$$

$$n x_n h(n) \xrightarrow{\mathcal{Z}} -z L_x'(z)$$